

Jess Jones

 Bristol, UK
  jess.jones@bristol.ac.uk
 07876 116588
  [jessj0nes.github.io](https://github.com/jessj0nes)
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Summary

PhD candidate at the University of Bristol (FARSCOPE CDT) with a First Class BSc in Computer Science. Research interests span deep learning, multi-agent reinforcement learning, and vision-based computing for robotics and trustworthy analysis of online content.

Education

PhD	University of Bristol , Robotics & Autonomous Systems	Bristol, UK
	- FARSCOPE Centre for Doctoral Training. - Supervisors: Prof. Sabine Hauert, OBE and Prof. Raul Santos-Rodriguez.	Sept 2023 – present
BSc	University of Bristol , Computer Science	Bristol, UK
(First Class)	Hargrave Prize for best final-year BSc thesis.	Sept 2020 – June 2023
Class	NAG HPC Student Prize (second highest accreditation).	
Honours	Netcraft Prize (top ten second-year cohort).	

Experience

CASM Technology , Developer & Researcher	Feb 2024 – present
Research and development of tools leveraging foundation models, deep learning, and other ML paradigms for analysis of social media, misinformation, and harmful content.	2 years 5 months
Motion Impossible , Full-stack Software Engineer	May 2023 – Sept 2023
- Designed the architecture and UI for simultaneous control of multiple robotic camera dollies. - Implemented communication network for one-to-one, one-to-many, and many-to-many device interactions.	5 months
University of Bristol — High Performance Computing Group , Research Intern	Bristol, UK
- Benchmarked MGCFD-OP2 across Arm architectures: ThunderX2, Graviton 2, Graviton 3. - Released mgcfd-op2 and babelstream packages to the Spack public repository.	June 2022 – Aug 2022
	3 months
CASM Technology , Developer & Researcher (summers)	June 2020 – Sept 2021
- Custom entity linking component in an automated fact-checking pipeline. - Research for UK Cabinet Office report on technologies for operational delivery.	1 year 4 months
Herb & Bloom , Owner & Director	2017 – 2019
Indoor vertical farming start-up. Responsible for all aspects of establishing, operating, and managing the business.	2 years
Cubitt House , Operations Director	2014 – 2017
- Five-site premium pub and hotel group with ~£14m turnover. - Operational delivery across sites, management performance, procurement, and strategic planning.	3 years

Publications

Assessing VLM-Driven Semantic Affordance Inference for Non-Humanoid Robot Morphologies 2026

Jess Jones, Raul Santos-Rodriguez, Sabine Hauert

[10.65109/WTKR8312](#) (Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS))

CoralLite: μ CT Reconstruction of Coral Colonies from Individual Corallites 2026

Jess Jones, Leonardo Bertini, Kenneth Johnson, Erica Hendy, Tilo Burghardt

(Proceedings of the CVBMC Workshop at the International Conference on Pattern Recognition (ICPR))

Awards

Hargrave Prize Aug 2023

- Best final-year undergraduate thesis on the BSc Computer Science programme at the University of Bristol.

NAG HPC Student Prize July 2023

- Second highest accreditation on the High Performance Computing programme (2023).

Netcraft Prize July 2022

- Awarded to the top ten second-year Computer Science students at the University of Bristol (2022).

Projects

MARL Framework for Swarm Communication Sept 2024 – present

- Simulation framework for efficient MARL experiments on Bristol swarm group's DOTS robots.
- Novel paint-mixing scenario testing communication accuracy and interpretation across agents.
- Evaluated a broad range of MARL algorithms and deep learning architectures.

3D Corallite Reconstruction (BSc thesis) Sept 2022 – May 2023

- UNet-ViT pipeline on volumetric micro-CT inputs; novel topological loss function.
- Fully automated Blender API system for interactive 3D structural analysis.

Lattice-Boltzmann HPC Sept 2022 – May 2023

- Distributed CFD optimisations with OpenMP, MPI, and OpenCL.
- Architecture-specific serial optimisations, multi-node scaling, GPU offloading.

Skills

Languages: C, C++, C#, Python, Java, Go, Haskell, SQL

Tools & Frameworks: PyTorch, TensorFlow, OpenMP, MPI, OpenCL, AWS, Git, Unity, Unreal Engine

References

Prof. Sabine Hauert, OBE

- PhD supervisor, University of Bristol — sabine.hauert@bristol.ac.uk

Prof. Raul Santos-Rodriguez

- PhD supervisor, University of Bristol — enrsr@bristol.ac.uk

David Weir

- Director, CASM Technology — davidw@casmtechnology.com